

Capstone Project Proposal

Team and Project Objective:

Project Id : Ind - 21 ; Group No : 20 ; Supervisor : Md. Ishrak Ahsan (ishrak@teacher.cse.buet.ac.bd)

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Organization and external mentor information and contact :

Organization: Samsung R&D, Bangladesh

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Domain overview and problem statement :

Title: Food and calorie intake tracking using phone

Tracking daily food and calorie intake is essential for a healthy lifestyle and weight management. Existing mobile applications typically require manual entry of food type, portion size, and calories, which is time-consuming and discourages consistent usage. ***This project aims to automate food logging using Vision AI, enabling users to capture a meal image and receive automated suggestions for food type, quantity, and estimated calories.***

High-level overview of scope and objectives :

The scope of this project is to develop a **smartphone-based food and calorie tracking system** that automates food logging using image recognition integrating AI model . The main objectives of the project are:

- To **reduce the effort** required for users to track their daily food intake
- To **automatically detect food type and estimate quantity** using Vision AI
- To **provide calorie estimates** based on recognized food items
- To **improve user engagement** by simplifying the food logging process
- To offer a **cost-effective alternative** to existing paid applications

Justification of why the project is complex:

This project is complex because it must be **fast and easy** to use while still providing accurate food recognition. **Food classification** is difficult due to the wide variety of foods and similar appearances. **Estimating portion size** from a single image adds further challenge.

Additionally, the project requires deep knowledge in multiple domains, including **Computer vision and deep learning, Image classification and object detection, Nutrition and calorie databases, Mobile application development**

Exploration of solution and implementation methods :

1. Initially, the user will be guided to capture food images from a predefined angle and posture to ensure consistency. Later can be upgraded.
2. The captured image will be processed using image **segmentation** techniques to separate individual food items.
3. **Portion size** will be estimated by calculating the segmented area and calibrating it using a real-world reference object, such as the user's thumb, for scale estimation.
4. A trained deep learning model will then **analyze the segmented food regions** to identify food types, estimate nutritional components, and compute total calorie intake.